PULSE

Engine Health Intelligence

DS-351 Data Visualization
Narrative Web Visualization — Project Report

Project URL: https://zainwajid775.github.io/DS-351-Project/

Team Members

Abdullah Yasin	2023049
Zain Wajid	2023775

Step 1: Data Analysis & Exploration

1.1 Dataset Overview

Dataset Name	Automotive Vehicles Engine Health Dataset
Source	Kaggle (Author: Parv Modi)
Records	19,535 engine sensor readings
Features	6 mechanical sensor variables + 1 binary condition label
Target Variable	Engine Condition: 1 = Good (63.1%) | 0 = Faulty (36.9%)
Missing Values	None — completely clean dataset
Theme	Real-time engine fault detection via sensor telemetry

The dataset represents a fleet of automotive engines captured in operation, each record corresponding to a single moment of sensor telemetry. The class distribution (63/37 Good/Faulty) is realistic and workable — not so skewed as to invalidate comparative analysis, yet clearly imbalanced in a way that reflects real-world fault detection scenarios.

Feature Descriptions

Feature	Unit	Description
Engine RPM	Rotations/min	Rotational speed of the engine. Faulty engines average ~20% higher RPM.
Lub Oil Pressure	Bar	Lubrication oil pressure. Critical to reduce friction; loss causes seizure.
Fuel Pressure	Bar	Fuel delivery pressure. Wide variance; extremes in both groups.
Coolant Pressure	Bar	Cooling system pressure. Outliers in either direction precede faults.
Lub Oil Temperature	°C	Lubricating oil temperature. Faulty engines slightly hotter (~0.6°C).
Coolant Temperature	°C	Engine coolant temperature. Healthy engines slightly warmer on average.

1.2 Research Questions & Motivation

Research Question 1

Q1 — Do Engine RPM and Fuel Pressure significantly differ between good and faulty engines?
Motivation: The Bosch Automotive Handbook documents RPM instability and fuel pressure irregularities as key precursors to valve train and injector failures. This question was additionally motivated by a personal experience — a family vehicle's intermittent stalling before complete engine failure, later diagnosed as a fuel pressure regulator fault.
External Evidence: SAE Technical Paper 2019-01-0232 (Fuel Injection Diagnostics) supports fuel pressure variance as a leading failure indicator in port and direct injection systems.

Research Question 2

Q2 — Do lubricant oil pressure and coolant pressure serve as reliable indicators of engine fault?
Motivation: SAE Paper 2019-01-0232 shows oil pressure loss causes engine seizure within seconds. Motivated by engineering coursework on thermodynamic systems and Haynes/Ford repair manuals documenting oil/coolant pressure as the two most common catastrophic failure precursors.
Key Hypothesis: The fault signature is not a simple threshold but a pattern of instability — multi-modal distributions across pressure features suggest multiple concurrent failure mechanisms (pump failure, leak, blockage).

Research Question 3

Q3 — How do lubricant oil temperature and coolant temperature interact to predict engine failure?
Motivation: ASME research demonstrates accelerated bearing wear above 95°C sustained coolant temperature. The compound thermal stress hypothesis — both oil and coolant temperature elevated simultaneously — represents the primary mechanical failure pathway.
Key Finding: The interaction heatmap confirms that compound thermal stress (high oil temp AND high coolant temp simultaneously) produces substantially worse fault rates than either variable alone.

1.3 Exploratory Data Analysis Results

Phase 1 — Distribution Analysis

Engine RPM exhibits a right-skewed distribution, consistent with the wide operational load range of automotive engines (400–2,000+ RPM). Temperature features cluster tightly near physiological norms with extreme-value outliers that represent critical operating conditions. Pressure features show more complex multi-modal distributions — particularly in faulty engines, where multi-modal peaks suggest multiple concurrent failure modes operating at different characteristic pressure regimes.

Phase 2 — Correlation Analysis

Sensor Pair	Correlation / Insight
Lub Oil Temp ↔ Coolant Temp	$r \approx 0.45$ (moderate) — shared thermal behavior from engine block.
RPM ↔ Pressure Variables	Weak correlation — RPM is largely independent of pressure circuits.
Cross-pressure variables	Near-zero — hydraulic circuits are physically separate and independent.
Multicollinearity check	No multicollinearity detected — each sensor adds independent diagnostic value.

Phase 3 — Q1 Findings: RPM & Fuel Pressure

- Faulty mean RPM: 885 vs. healthy mean RPM: 736 — a statistically meaningful +148 RPM difference.
- Engines operating above 1,400 RPM show the highest concentration of fault-classified readings.
- Fuel pressure extremes — both low (starvation mode) AND high (regulator failure mode) — are fault-associated.
- RPM alone cannot diagnose a fault due to significant distribution overlap, but combined with other signals it becomes a powerful discriminating feature.

Phase 4 — Q2 Findings: Oil & Coolant Pressure

- Pressure INSTABILITY — not a single low-pressure threshold — is the key fault signature.
- Multi-modal distributions in faulty engines reveal at least three concurrent failure modes: pump failure, active leak, and blockage.
- Healthy engines maintain tighter, more unimodal pressure distributions within expected operating ranges.
- Mean pressure comparisons between classes are less informative than distributional shape analysis.

Phase 5 — Q3 Findings: Thermal Interaction

- Extreme temperatures elevate fault rates individually, but the compound effect is disproportionately worse.
- The interaction heatmap (oil temp × coolant temp cross-tabulation) confirms: high oil temp AND high coolant temp simultaneously produce the worst fault outcomes.
- Mean temperature differences between classes are small because faulty engine temperatures include both extremes (hot AND cold spikes), which partially cancel when averaged — making distributional analysis far more informative than mean comparison.
- Sustained coolant temperatures above ~88°C combined with oil temperatures above ~84°C represent the most dangerous operational state.

Phase 6 — Cross-Feature Summary

Signal Type	Feature(s)	Finding
Strongest fault signal	Engine RPM	+148.7 RPM mean difference; RPM >1400 is highest-risk zone.
Most diagnostically stable	Temperatures (oil & coolant)	Small mean delta; distributional extremes matter more.
Richest failure signature	Pressure variables	Multi-modal instability reveals multiple failure mechanisms.
Compound predictor	Oil Temp × Coolant Temp	Interaction heatmap confirms dual thermal stress pathway.

Step 2: Storyboarding

2.1 Narrative Arc & Story Structure

The project follows a classic journalistic narrative arc — opening with a compelling hook, introducing the world (the dataset), diving into author-driven discoveries chapter by chapter, then handing control to the reader for open-ended exploration, and finally delivering a synthesis of findings.

Act	Purpose
Act I — Hero / Hook	Establish the stakes: 1 in 3 engines in this fleet is faulty. Something is wrong.
Act II — The World (Ch. 01)	Introduce the dataset: 19,535 readings, 6 sensors, the class split.
Act III — Discovery (Ch. 02–03)	Author-driven scrollytelling revealing each diagnostic pattern progressively.
Act IV — Exploration (Ch. 04)	Reader-driven: interactive scatter plot with axis/filter controls.
Act V — Conclusion (Ch. 05)	Synthesize the three research questions into a cohesive diagnostic model.

2.2 Chapter-by-Chapter Storyboard

Hero Section — 'Something is Wrong'

Storyboard: Full-screen cinematic entry with animated rotating ring motif
Visual: Dark background (#060a0f), large geometric ring animation (CSS, infinite rotation), subtle grid overlay at 6% opacity.
Typography: Large display heading 'ENGINE HEALTH INTEL' with key stat callouts — 19,535 readings, 63.1% healthy (green), 36.9% faulty (red).
Interaction: Blinking red dot in nav-bar (live-system aesthetic). Scroll-down hint arrow with bounce animation.
Purpose: Establish urgency and a 'mission control' visual metaphor. The reader should feel they are looking at real telemetry.
Static element: Hero statistics (19K, 63.1%, 36.9%) appear immediately on load without user interaction.

Chapter 01 — 'The Fleet at a Glance'

Storyboard: Feature cards + two D3 charts side by side
Left content: Six feature-cards, each with a color-coded icon (emoji), sensor name, and one key insight bullet.
Right charts (2-column grid): (a) Donut chart — class distribution pie with hover tooltips. (b) Bar comparison chart — normalized mean values for all 6 sensors, Good vs Faulty, color-coded green/red.
Interaction: Hover tooltips on both charts revealing exact values.
Transition: Cards fade in on scroll (IntersectionObserver). Charts render via D3 on section entry.
Purpose: Author-driven — establish the vocabulary and overview before going deep.

Chapter 02 — 'RPM: The Loudest Signal' (Scrollytelling)

Storyboard: Sticky visualization + narrative steps — 4-step scrollytelling sequence
Step 1: Show all 19,535 readings as a grey histogram. Neutral framing: 'looks like a normal distribution.'
Step 2: Filter to healthy engines only. Distribution shifts left, peaks at 700–750 RPM (green bars).
Step 3: Filter to faulty engines only. Distribution shifts right, peaks at 900–950 RPM (red bars). +148 RPM difference annotation appears.
Step 4: Both overlaid — overlap highlighted, with editorial note that RPM alone is insufficient.
Mechanism: Intersection Observer on narrative steps triggers D3 animated transitions (smooth bar height tweens). Left panel is sticky (position: sticky), right panel scrolls past.
Key design choice: The animation IS the insight — seeing the distribution shift reveals the pattern in a way a static chart cannot.

Chapter 03 — 'Reading the Warning Signs'

Storyboard: Gauge grid + scatter plot + box plots
Gauges (reader entry point): 6 circular gauges, one per sensor, showing mean values for healthy vs faulty engines with arc animations on load.
Scatter Plot: 200-point random sample, RPM (x-axis) vs Fuel Pressure (y-axis), colored by condition (green/red). Hover tooltip shows all 6 sensor values + condition for each point.
Box Plots (2-column): (a) Fuel Pressure — Good vs Faulty with Q1/Q3/median/whiskers. (b) Lub Oil Pressure — same layout. These visualize distributional instability, not just mean differences.
Interaction: All charts interactive — tooltips on scatter points, hover on box elements.
Purpose: Demonstrate the multi-sensor diagnostic picture before handing control to reader.

Chapter 04 — 'Your Turn to Explore'

Storyboard: Reader-driven interactive scatter plot with controls
Controls panel: X-axis selector (all 6 sensors), Y-axis selector (all 6 sensors), Condition filter (All / Healthy / Faulty).
Chart: D3 scatter plot that re-renders smoothly on any control change, using transition animations.
Points: Color-coded green (healthy) / red (faulty) with hover tooltip.
Purpose: Transition from author-driven to reader-driven. The reader can now test their own hypotheses — e.g., 'Does oil pressure vs. coolant temp show separation?'
Design note: Default axes set to RPM (x) vs Fuel Pressure (y) as a natural continuation from Ch. 02.

Chapter 05 — 'What the Data Tells Us'

Storyboard: Three conclusion cards + overall summary
Layout: Three cards side by side, one per research question (Q1, Q2, Q3), each with a colored tag, headline finding, and 2-sentence support.
Visual: Clean, minimal card style — white background, accent color border. No charts — words are the visual here.
Closing paragraph: Synthesis of all three questions into an actionable diagnostic model.
Acknowledgements: Team names, registration numbers, dataset credit (Parv Modi / Kaggle).
Final design choice: Deliberately sparse — after 4 chapters of dense visualization, the conclusion breathes.

2.3 Navigation & Flow Design

- Fixed top navigation bar with chapter anchor links (Overview, Patterns, Signals, Explorer, Findings).
- Smooth scroll behavior via CSS scroll-behavior: smooth.
- Chapter labels use Space Mono monospace for a 'technical readout' aesthetic.
- Progress: No explicit progress bar — scroll position is communicated by sticky chapter labels.

2.4 Interaction Techniques Summary

Technique	Chapter	Purpose
Scroll-triggered animation	Ch. 02 (RPM)	IntersectionObserver activates narrative steps; D3 histogram transitions.
Sticky visualization	Ch. 02 (RPM)	Left narrative scrolls, right chart stays fixed — classic scrollytelling.
Hover tooltips	Ch. 01, 03, 04	Reveal exact sensor values on demand without cluttering the base chart.

Donut chart interaction	Ch. 01	Click/hover reveals class percentages and counts.
Axis/filter selectors	Ch. 04 (Explorer)	Reader-driven hypothesis testing via dropdown menus.
Gauge animations	Ch. 03	Animated arc fill on section entry creates engagement.
Fade-in on scroll	All chapters	IntersectionObserver adds .visible class; CSS transitions fade elements in.

Step 3: Visual Design Study

3.1 Visual Concept & Intended Audience

The intended audience is technically literate but not necessarily specialized in data science — engineering students, junior automotive technicians, and curious general readers who appreciate precision and clarity. The visual language must communicate authority (data-backed, technical), urgency (faults are real and consequential), and accessibility (the story should unfold without requiring a statistics background).

The guiding metaphor is a mission control interface — inspired by aerospace telemetry dashboards, medical monitoring systems, and industrial IoT displays. The palette, typography, and motion design all reinforce this metaphor.

3.2 Color Palette

Role	Hex Value	Rationale & Usage
Background (Primary)	#060a0f	Near-black with a deep blue undertone — evokes a dark cockpit/control room.
Surface (Cards)	#0d1520	Slightly lighter — creates depth hierarchy between bg and content areas.
Healthy (Signal Green)	#00e5a0	Vivid teal-green: unmistakably positive, high-contrast against dark bg. Used for healthy engine class, positive metrics, nav logo.
Faulty (Signal Red)	#ff3d5a	Hot coral-red: immediately communicates danger/fault. Used for faulty class, warning indicators, blinking nav dot.
Accent (Amber)	#f5a623	Warm amber for secondary highlights and callout text — adds warmth without competing with primary signals.
Body Text	#e8edf5	Soft warm white — readable without the harshness of pure white on dark bg.
Muted Text	#5a6a80	Mid-grey-blue for secondary information, labels, navigation links.

Visual inspiration: Bloomberg Terminal (dark dense data display), NASA Mission Control photography, Grafana monitoring dashboards. The key design principle is high-information-density dark UI, where color carries semantic meaning rather than decorative purpose.

3.3 Typography

Font / Role	Rationale
Syne (display, headings, nav logo)	Geometric sans-serif with distinctive character — conveys modernity and precision. The bold 800 weight creates strong visual hierarchy. Used for hero heading, chapter titles, card headers.

Space Mono (chapter labels, tags)	Monospaced typeface with a 'code terminal' aesthetic. Reinforces the technical / telemetry metaphor. Used for chapter labels (e.g., 'Chapter 01 / Dataset Overview') and section tags.
Inter (body text, chart labels)	Clean neutral sans-serif with excellent legibility at small sizes. Standard for technical documentation and data interfaces. Used for all body copy, descriptions, tooltip text.

Font pairing rationale: The three-font system creates three distinct registers — Syne for narrative identity (this is a story), Space Mono for system identity (this is technical), Inter for content readability (this is information). Each font activates at a different cognitive level.

3.4 Layout & Spacing Principles

- Full-viewport sections (min-height: 100vh) for the hero — establishes cinematic entry.
- 7rem top/bottom padding on all content sections — generous breathing room prevents cognitive overload.
- 5vw horizontal padding — fluid and responsive, scales naturally from desktop to tablet.
- Grid systems: 2-column grid for parallel comparisons (donut + bar chart; two box plots); single-column for narrative text.
- Sticky layout (Ch. 02) breaks the rigid grid intentionally — signals a mode shift to scrollytelling.

3.5 Motion & Animation Design

Animation	Technique & Purpose
Hero ring rotation	CSS transform rotate (40s linear infinite) — creates a living, breathing landing that signals the data is 'active.'
Navigation dot blink	CSS opacity keyframe (1.4s) — reinforces the 'live monitoring' aesthetic subtly.
Scroll-down arrow bounce	CSS translateY keyframe (2s) — standard affordance cue for scroll interaction.
Section fade-in	IntersectionObserver + CSS opacity/translateY transition (0.6s ease-out) — elements appear as user arrives at them, reducing initial visual noise.
D3 histogram transitions	D3 selection.transition().duration(600) — smooth bar height animation when RPM filter changes, making the distribution shift legible.
Gauge arc fill	D3 arc tween on section entry — provides a clear visual summary before the scatter/box charts.

3.6 Visual Inspiration Sources

Reference Visual Media Surveyed
The Pudding (pudding.cool) — Benchmark for scrollytelling narrative flow. Their 'Film Dialogue by Gender' and 'Music Map' pieces informed the sticky-left / scrolling-right layout architecture.
New York Times Visual Investigations — Dark-theme editorial data pieces. Their use of high-contrast color coding (red/green, red/blue) for categorical distinctions directly influenced the healthy/faulty palette choice.
Bloomberg Terminal UI — Dense, dark, grid-based data display. Influenced the overall dark background palette and the decision to prioritize information density over whitespace.
NASA Mission Control photography — The green-on-dark monochrome aesthetic of CRT telemetry displays inspired the --healthy green (#00e5a0) and the overall 'monitoring station' metaphor.
Grafana / Datadog dashboards — Modern industrial monitoring interfaces. Influenced the gauge visualization design and the blinking status indicator in the navigation bar.
Graphic novel 'Watchmen' (Alan Moore, 1986) — Non-linear narrative structure and panel-within-panel layout inspired the idea that a data story can use visual rhythm to control pacing, not just text.

3.7 Design Decisions Summary

Decision	Alternative Considered	Rationale for Choice
Dark background theme	Light / white background	Dark theme fits the 'telemetry monitoring' metaphor; also reduces glare fatigue when readers engage for extended periods.
3-font system	2-font system	Space Mono for labels creates a uniquely technical register that neither Syne nor Inter could provide alone.
Signal green + signal red	Blue/orange or blue/red	Green = healthy, red = danger is the most universally understood signal pair in engineering and medical contexts.
Sticky scrollytelling for Ch. 02	Paginated slides or tabs	Scrollytelling provides continuous narrative flow without forcing click interactions; more immersive.
200-point sample for scatter	Full 19,535 points	Full dataset overplotting would obscure individual points; 200 representative samples preserve pattern while maintaining legibility.
Chapter numbering (01, 02...)	Descriptive section titles only	Numbering creates a sense of structured progression; reinforces the 'mission briefing' metaphor.